
End to End Active Learning Framework for chest-abdominal CT Scans Segmentation

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ABSTRACT State-of-the art deep learning approaches for segmentation require large, high-quality annotated datasets for training and evaluation. However, the manual development cycles involved in this process are time-consuming. Moreover, the data labeling can be prohibitively expensive, particularly in medical imaging, where specialized radiological expertise is required to accurately annotate organs or structures of interest in complex imaging modalities such as chest-abdominal CT scans. In this work we utilize an active learning (AL) strategy in order to achieve high segmentation model performance with a low annotation budget. The key idea is to leverage the power of machine learning to actively select informative samples from the unlabeled data for annotation, instead of relying solely on random or heuristic-based sampling. This enables us to significantly reduce the amount of labeled data needed for training, while still maintaining high segmentation accuracy. Our contributions include, creation and automation of end-to-end (E2E) machine learning flow, applied to solve organ binary segmentation tasks in chest-abdominal CT scans using AL. Additionally, We have established an active learning infrastructure that supports valuable case selection, effectively reducing the annotation workload in medical imaging. This dynamic system allows for easy adjustments to accommodate new segmentation tasks, datasets encompassing different anatomies, and other parameters within the active learning flow, that reduces the annotation burden in medical imaging and may be used for other application domains. Finally, we demonstrate the tangible benefits of AL approach in terms of reduction in annotation cost and training time by the AL flow for a given supervised learning medical segmentation task. Our implementation achieved a 0.908 mean dice score [0.895, 0.921] and 0.833 mIoU [0.816, 0.849] for spleen segmentation task and 0.855 mean dice score [0.848, 0.861] and 0.785 mIoU [0.777, 0.792] in liver segmentation task. This work provides a solution to the challenges of time-consuming and expensive data annotation in the current ML-DL era, particularly in medical imaging and other computer vision applications involving large data volumes.

INDEX TERMS Artificial intelligence, Deep Neural Networks, Semi-supervised learning, Active Learning, CT organ Segmentation

1. INTRODUCTION

Throughout the last decades, research in machine learning has advanced the automation and analysis of medical images to support diagnosis, treatment planning and treatment delivery workflows in the healthcare domain (Zendel et al. 2017; Zhang et al. 2020; Varoquaux and Cheplygina 2022). Image segmentation is a key component in such systems, involving partitioning images into multiple meaningful objects (Long et al. 2015). Abdominal organ segmentation in Computed Tomography (CT) is a key component in computer-assisted diagnosis (CAD) applications, including surgery navigation, visual augmentation, radiation therapy and biomarker measurement systems for abdomen diagnosis (Isensee et al. 2019; Zhang et al. 2020).

In general Deep Learning Segmentation models require substantial datasets for training. In the medical domain,

availability of high-quality labeled data is even more cumbersome, in order to reach the high level of accuracy requirements (Beluch et al. 2018). Organ segmentation in CT is challenging especially in situations where the testing sets have unseen or severe diseases and low image quality, heterogeneous lesions and noise, and in cases of generalization to new institutions with different data distribution from the original training set. The cost of time consuming labeling process with the size of datasets constantly growing, is even more challenging in the medical domain, where there is the need for professional expertise with familiarity with the specific clinical domain, complex medical data, regulatory compliance issues and trust in trained models. (Tajbakhsh et al. 2020).

Numerous image segmentation algorithms have been proposed in the computer vision and machine learning

literature, including classic computer vision approaches as well as deep learning (DL) approaches in the last decade, which have yielded the highest accuracy rates for most benchmarks. Surveys list common approaches including fully convolutional networks (Long et al. 2015), encoder-decoder based approaches (Cho et al. 2014; Badrinarayanan et al. 2017; Chen et al. 2021), RCNN based models (He et al., 2017), the DeepLab family models (Chen et al. 2018) and recently even promptable segmentation (Kirillov et al. 2023) U-net is an image segmentation classic architecture that has become successful in medical image analysis. It provides good performance on various medical image segmentation tasks using a scarce amount of training data. This is important in the medical imaging community as properly labeled images are often limited (Chen et al. 2018; Nahian et al. 2021).

3D U-net is an extension of the basic U-net framework that enables 3D volumetric segmentation which is CT scans where volumetric information can be important. But will be a heavier model and will suffer from long computation time and large input files (full CT scan). (Ronneberger et al. 2015; Isensee et al. 2020)

Current Deep learning approaches rely on training on large datasets. For semantic or instance segmentation this requires pixel-wise labels that are time consuming and expensive to annotate. (Hung et al, 2018; Tao et al, 2020).

Several strategies have emerged in this field that seek to reduce the amount of labeled data required to achieve acceptable predictive performance, including semi-supervised (Cho et al. 2014), transfer learning (Dmitriev and Kaufman 2019), active learning (AL) (Beluch, 2018; Laradji et al. 2020; Agarwal et al. 2021) and very recently, based on Self Supervised Learning (SSL) (Chen, 2020). Active learning and semi-supervised learning utilize unlabeled data in two different ways. Active learning selects the most valuable unlabeled data points and submits it to the expert system for labeling. Semi-supervised learning establishes the connection between unlabeled data and labeled data, uses the inherent information of unlabeled data to improve the model, and assigns pseudo-labels to those samples with high confidence (Ren et al. 2022). Transfer learning utilizes existing labeled data from one task to help learn related tasks for which limited labeled data are available (Kale and Liu, 2013). Big self supervised models are strong semi supervised learners SSL requires positive semantic pairs, this is commonly handled by applying data augmentation techniques to the same input. Active learning can be used to further generalize SSL and build more sophisticated positive views (Chen et al, 2020, Cabannes et al, 2023).

Active learning (AL) considers a dynamic scenario in which a model can iteratively select the most informative labels for the samples that it believes will be most useful for solving a given task (Kim et al. 2020; Shin et al. 2021). This is in contrast to passive supervised learning in which the model is focused at learning a mapping from a fixed set of input-output pairs and all data is labeled. (Laradji et al. 2020; Shin et al. 2021). The key idea is to achieve higher accuracy with fewer labeled instances when compared to conventional supervised learning methods. This is accomplished by allowing the Active Learner to choose the data it wants to learn from. Thus, the Active Learner poses queries in the form of unlabelled data instances and requests the oracle (i.e., a domain expert or any other labeling source) to provide corresponding labels (Agarwal, 2021).

Active learning is an iterative process, that in each iteration a selection strategy selects a set of unlabelled instances, the instances selected are labeled by the annotator, the instances are added to the training set, and the learning algorithm is trained with the new training set (Reyes et al. 2018)

Therefore, Active learning (AL) which proposes a dynamic scenario, in which a model can iteratively select the most informative labels for the samples, namely those that will be most useful for solving a given task, can effectively reduce the annotation cost for training a semantic segmentation model. (Laradji et al. 2020; Budd et al. 2021).

The objective of this research is to suggest an automated infrastructure using active learning that will be deployed on chest-abdominal organ CT segmentation tasks and implementation on the liver-spleen segmentation use case. We focus on creating and automating the machine learning workflow using the data tools used in industry and evaluate whether active learning can improve the accuracy of segmentation together with reducing the annotation costs. To demonstrate the power of our approach, we used public datasets for different image segmentation tasks, which include pixel-level labels of 12 organs including spleen and liver. This work focuses on creating infrastructure that uses active learning to optimize the labeling process and in particular reduce the amount of labeled data in the context of CT abdomen organ segmentation in medical imaging. In this section we review the most relevant literature for AL and medical image segmentation.

We summarize our key contributions:

- We design a novel automatic machine learning flow suitable for binary image segmentation. The flow manages the end-to-end (E2E) image segmentation

task: preprocessing, training, evaluation and refinement loops that involve making predictions on the available dataset and selecting a subset of data that will be most beneficial for the next training iteration. We show how such a flow can be applied to SOTA supervised segmentation of abdominal CT. Our active learning approach achieves accuracy that is on par with, and in some instances surpasses, fully supervised baselines. Notably, our method achieves these results while requiring less annotated data than traditional fully supervised methods. (Section 3.1).

- We have established an active learning infrastructure that supports valuable case selection, effectively reducing the annotation workload in medical imaging. This dynamic system allows for easy adjustments to accommodate new segmentation tasks, datasets encompassing different anatomies, and other parameters within the active learning flow. (Section 3.3.2, 4.3).
- We conduct a comprehensive empirical analysis to assess the efficiency of our methodology, specifically quantifying the reduction in annotation cost and training duration attributable to our approach. (Section 5).

In sum, we demonstrate that leveraging AL can be extremely powerful for complex grayscale pixel-wise tasks. As the problem of GT (ground truth) annotation for deep learning is a fundamental problem, we expect this work to yield a significant contribution in many other domains.

II. PRIOR WORK

Several algorithms and workflow have been developed to address the task of image analysis and organ segmentation. For reference, Chen, S. et al. (2022) research proposes a new learning-based CRF (Conditional Random Fields) approach for image segmentation that allows the CRF to use features learned by a CNN, optimizing the CRF and CNN parameters concurrently. The method was evaluated in three segmentation applications: aorta and pulmonary artery segmentation in non-contrast CT, white matter hyperintensities segmentation in multi-sequence MRI and ischemic stroke lesion segmentation. Dataset for CT included 25 non-contrast lung scans. The model implementation was using 3D UNet as the basic CNN architecture.

Gende, M. et al. (2022). proposed a fully-automated end-to-end architecture to solve simultaneous segmentation and screening of epiretinal membrane (ERM) signs in Optical Coherence Tomography images. Three different approaches were tested using different encoder configurations to compare their accuracy and efficiency in producing objective and accurate segmentation and screening of the ERM. The models were trained to extract and select relevant visual features without the need for manual feature selection or fine-tuning intermediate steps. This approach simplifies the development of CAD systems and provides significant advantages during training and inference over existing methods for ERM segmentation. This study uses 20 OCT volumes - 8 of them present ERM, while the remaining 12 belong to healthy eyes. model architecture based on the Feature Pyramid Network (FPN).

In Hussain, M. et al. (2022) work, highlights the challenges of obtaining a large labeled dataset for COVID-19 pneumonia infection segmentation due to the limited availability of annotated data. To overcome that challenge an active learning with sample re-weighting-based on semi-supervised learning case selection strategy is proposed. enabling the selection of reliable and more informative samples from machine-annotated batch data. The weighting function factorizes the similarity of the gradient descent directions to those of expert-annotated data, and the gradient magnitude of the last layer of the deep model.

In this research, we utilize the power of Active Learning and design the framework to support organ segmentation using active learning and Unet architecture. As opposed to previous work described in the related work section we decided to separate the active learning optimization flows parameter selection (Active Learning flow) from the training algorithm, this will allow the user gain more flexibility in the design of the active learning pipeline, allows us to experiment with different parameter selection strategies and choose the most effective one for the given task. Moreover, Efficiency parameter selection step can be performed on a smaller subset of the data, reducing the computational cost and training time.

The remainder of the paper is structured as follows: section 3, introduces the methods, outline of the framework and E2E flow. Section 4, describes the implementation details and experiments. Section 5, shows our results describing the performance of the suggested framework in comparison to previous work. Section 6, summarizes with our conclusion and future work.

III. METHODS

In this section, we describe the resources that are used for this research and framework definition. Outline of framework describe the training and refinement loops and preprocessing flow (subsection 3.1), Network architecture (subsection 3.2.1), Evaluation metrics (subsection 3.2.2), outline of refinement flow using Active Learning (subsection 3.3), Detailed explanation of framework components (subsection 3.3.1), Heuristics Acquisition functions (subsection 3.3.2) and implementation details and experiments set to define the optimal solution for organ segmentation using active learning (Section 4)

For this work two open source third party datasets were utilized. The Medical Segmentation Decathlon (MSD)(Antonelli et al 2021) contains 131 CT with liver,

- ii. Spleen: The data set consists of 61 portal venous phase CT scans. The corresponding target ROI was the spleen.

3.1. Outline of Active Learning System:

Figure 1 presents the active learning flow, we begin with training an initial model based on randomly selected small subset of images, then we run an inference step on all unlabeled cases and start with refinement iterations. In each iteration we select a subset batch of images (based on an acquisition function) and submit them to the human annotator as a tagging assignment. The annotator will agree / fix all regions of interest, and the results will be aggregated and then fed into the segmentation model which will produce segmentation masks and probability maps.

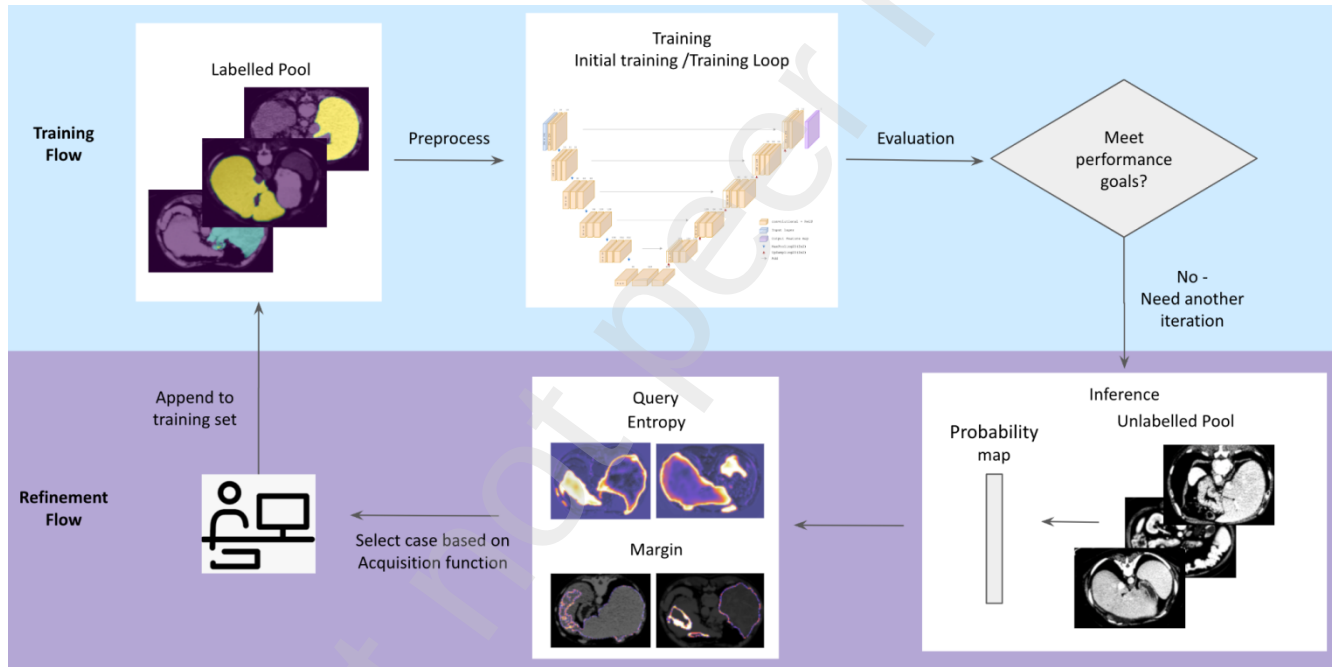


FIGURE 1. Training Flow refinement loop, explained in steps below

cancer and background labels and 41 CT scans with spleen and background labels and Synapse (Xu, 2016), contains 30 abdominal CT scans which contain 12 organs including spleen and liver. All images were de-identified and reformatted to the Neuroimaging Informatics Technology Initiative (NIFTI) format. Among the organs that were mapped, for this project we focus on spleen and liver. The dataset included the following scans:

- i. Liver: The data set consists of 201 contrast-enhanced CT images. The corresponding target ROIs were the segmentation of the liver and tumors inside the liver.

Finally, we evaluate the model's segmentation results on a held-out set, a set of images which the model has not encountered before. In case the model results reach the performance goals over on the held-out set, or we didn't see any improvement in the last k ($k=3$ in our experiment) iterations, we deploy the model and complete the training and refinement loops. Otherwise we start another iteration running inference with the newly updated model and trigger a new iteration.

3.2 Training and Evaluation Flow

end-to-end process responsible for training a new segmentation model using a subset of annotated images and generating a score to assess the model's performance.

3.2.1. DNN Model Architecture

We trained two Unet networks for each anatomy, one for spleen and one for liver. Performance was evaluated and the best model was selected. We also aggregated the best score in order to create a unified multi organ model.

As an Initial model we trained a 2D Unet. In each training iteration the network was fed with 2D slice volumes of size (256 x 256 x 1). The initial dataset was selected randomly by scan level, we used only slices that contained one of the organs, spleen and liver.

Our implementation is based on a traditional Unet architecture which consists of a contracting (left side) and an expansive (right side) path. The contracting path consists of the six repeated applications of two 3x3 convolutions with zero padding convolutions and with filter size 2^i ; $i=3..9$, each followed by a rectified linear unit (ReLU) and a 2x2 max pooling operation with stride 2 for downsampling.

At each downsampling step we double the number of feature channels. Every step in the expansive path consists of an upsampling of the feature map followed by a 2x2 convolution ("up-convolution") that halves the number of feature channels, a concatenation with the correspondingly cropped feature map from the contracting path, and two 3x3 convolutions, each followed by a ReLU. At the final layer a 1x1 convolution is used to map each 64- component feature vector to the desired number of classes. In total the network has 29 convolutional layers.

The model was implemented using a Keras library with a TensorFlow backend. For each anatomy we trained the Unet model from scratch using an Adam optimizer with learning rate of 0.001, binary cross entropy loss. Early stopping was configured after 4 patience with checkpoints to capture max mIoU score.

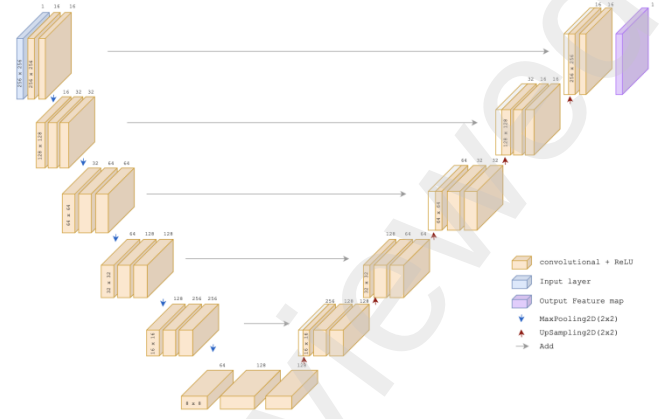


FIGURE 2. The Unet architecture, based on the implementation of Ronneberger et al. 2015. The input layer corresponds to the blue box. The output feature map is mapped to the purple box top right edge. Each yellow box corresponds to a multi-channel feature map. The number of channels is denoted on top of the box. The x-y-size is provided at the lower left edge of the box. The white boxes represent copied featured maps. The arrows denote the different operations.

3.2.2. Evaluation Metrics

In order to estimate the model improvement and to evaluate the performance two common metrics was selected mean IoU and Dice coefficient (F1 Score)

mIoU - The Intersection-Over-Union (IoU), also known as the Jaccard Index - mean IoU of the image is calculated by taking the IoU of all classes and averaging them (Tiu, 2019)

$$IoU = \frac{TP}{TP + FP + FN}$$

Equation 1 : IoU calculation

Dice Coefficient(F1 Score) - twice the Area of Overlap divided by the total number of pixels in both images) (Cabello, 2021)

$$Dice(F1) = \frac{2TP}{2TP + FP + FN}$$

Equation 2 : Dice(F1) calculation

Differences between those metrics can be found in calculation of mistakes made by the network; The IoU metric tends to have a "squaring" effect on the errors relative to the Dice score. So the Dice score tends to measure something closer to average performance, while the IoU score measures something closer to the worst case performance. (Ilyas et al., 2020).

In order to avoid edge cases where $(TP + FP + FN) = 0$ or where there is no true labeled image we added a smooth parameter approximately 0 for both numerator and denominator that will simplify our calculations.

Previous work with same implementation showed 93.43 dice score for Liver segmentation and 86.67 for spleen

segmentation on Synapses dataset Based on related work that was done using same framework with segmentation task, expected mean dice results: 88.80 (Amyar et al., 2020)

3.2.3. Initial Training Model (Iteration 0)

To provide a starting point for the active learning training and refinement loop an initial model was implemented. Dataset was randomly selected annotated scans of spleen and liver as initial training pools.

Data analysis - when reviewing the scans we noticed that only ~20% of the slices included masks. We decided that for training the initial baseline model we will use only images that were labeled with a mask, namely data with no annotations was dropped.

The working points (threshold) for the spleen and liver models were selected based on the validation set results and were set to 0.65, 0.49 respectively. The selection curves are visualized in Figure 3

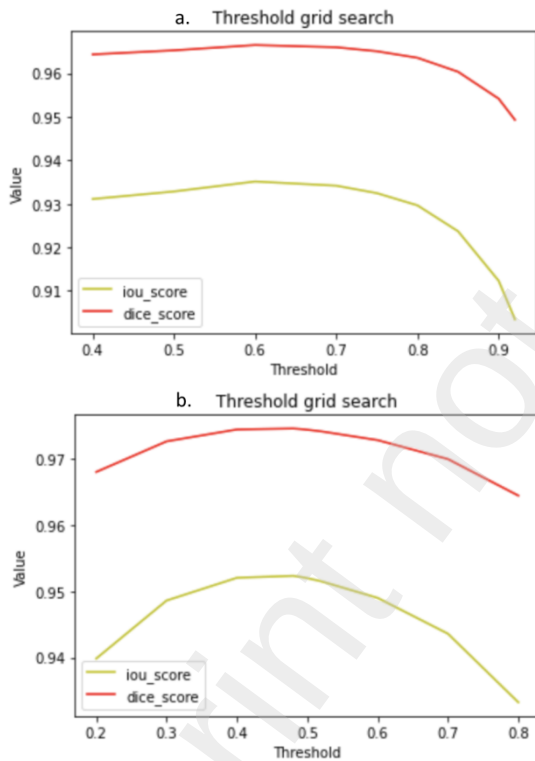


FIGURE 3. Performance curves on the validation set (plot a for spleen and plot b for liver). mIoU (yellow) and dice score (red) for each threshold, the selected threshold is one that maximizes the mIoU.

In each iteration we selected the working points which optimized the mIoU on validation set and evaluated performance on the held-out set.

3.3. Refinement Flow

The refinement flow aims to improve the performance of the model by selectively choosing the pixels that are most informative for the model's learning. By iteratively selecting the most informative pixels and adding them to the training set, the model is expected to gradually improve its performance with fewer labeled instances compared to traditional supervised learning. we will review the components

3.3.1. Active Learning Cycles

Case selection was done based on the heuristic acquisition function results (defined in section 4.3.2). In each iteration we ran the model as inference, calculated the entropy and margin on all available training images that we haven't trained on. The scores were sorted based on function results and random samples were selected from top scores added to the training dataset for the next training iteration. We continued to run active training loop selection and refinement iteration until our model met the performance goals on a full held-out set or we didn't see any improvement in the last three iterations.

Figure 4, shows the changes in prediction of the model, we can see that for random selection (middle row - most left image - iteration 0) the model is predicting badly, after one iteration, we can see that the area of segmentation is much more representative to label. After 5 iterations of model refinement the model manages to predict a single anatomy with the shape and area of the liver (bottom row - most right image- iteration 5). All parameters was selected based on experiments which will be presented in Experiment section (Section 5)

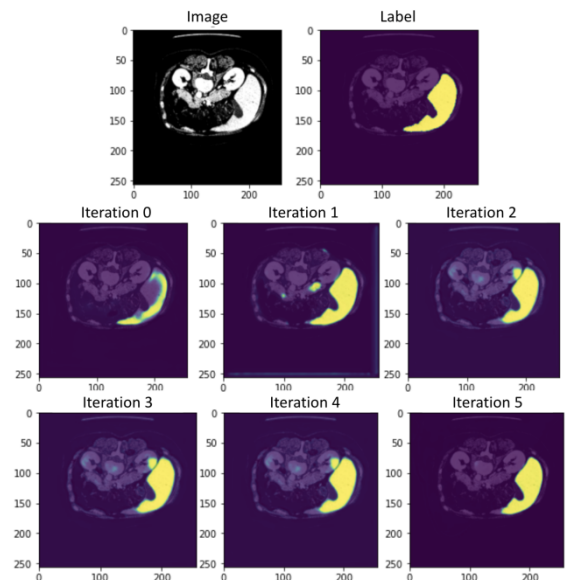


FIGURE 4. Demonstrating the implementation on liver segmentation, showing improvement of AL refinement iteration. The top row shows the original image and GT labeling from the held-out set. The bottom row shows the prediction results of the model in each AL refinement iteration. Where the leftmost corresponds to the prediction based on the initial model - cases selected randomly whereas the rightmost bottom image corresponds to the best result with selected model.

3.3.2. Heuristics Acquisition functions

In this project we compare different approaches for case selection based on the framework of uncertainty sampling. We select the image regions that will be the most informative and beneficial to our model. We will take into account uncertainty and diversity sampling parameters.

While more sophisticated strategies for case selection that were considered within our framework, margin and entropy score strategy proved to be effective, simple to implement and optimize run time of refinement loop. Therefore we decided to focus on these heuristics as part of this work.

Highest Margin score strategy- Case selection based on margin score was calculated as the sum of pixels with the highest uncertainty. We define the uncertainty value and count all pixels in the image within this interval. Where C represents the number of classes, N the number of pixels within the image, $\mathcal{E}$ the accepted interval, $p_{i,j}$ the predicted probability of each pixel belonging to the object of interest output by the model, which is obtained from the output of the segmentation model and threshold as the selected working point corresponding to each class. Justification for $\mathcal{E}$ selection can be found in the experiment section. In figure 5 we can see that areas that do not match the GT label were marked in red as the pixels with highest uncertainty.

$$\text{margin score} = \sum_{i=0}^C \sum_{j=0}^N |(p_{i,j} - \text{threshold}_i)| < \mathcal{E},$$

Equation 3 : General margin score definition ; denote: C - number of classes, N - number of pixels within the image; $P_{i,j}$ - predicted probability for specific class and instance, threshold_i - selected working point for specific class and $\mathcal{E}$ -accepted interval

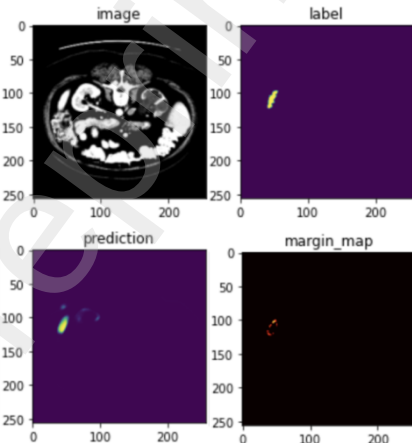


FIGURE 5. Illustration of the margin score based on visualization of the spleen data set. The bottom right image presents the pixels with the highest margin score.

Entropy Sampling strategy - Case selection based on highest mean entropy score, cross entropy score was calculated for each class then avg in order to create score per image. In figure 6 we can see that areas that are not matched to the label received the highest entropy score and were selected as regions of interest where the model is least certain about. Conditional entropy is calculate as:

$$\text{Cross entropy score} = - \sum_{i=0}^C p_i \log(p_i)$$

Equation 5: Formulation of our implementation for Cross Entropy Score based on Korner and Orlitsky, 1998 ; Denote: C = number of classes and p =predicted probability

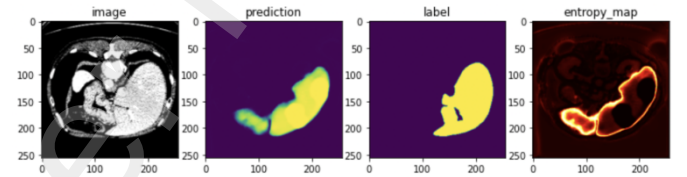


FIGURE 6. Illustration of the entropy score on a sample of liver data. The rightmost image presents the pixels with the highest entropy.

IV. EXPERIMENTS AND PARAMETERS OPTIMIZATION

In this section we will describe the flow implementation and experiments conducted to optimize flow parameters. We will cover the dataset selection, preprocessing flow, model implementation and active learning parameters optimization

4.1. Data

Data scans were splitted to three groups, a Training, Validation and Test set (Held-Out. The test set does not participate in the training / tuning processes. In every training iteration we evaluated model improvements based on this Test set.

Data was transposed from 3D NiftI files to 2D images and was tupled into imagekey-label format. For synapse dataset which are labeled with multi-organs, the segmentation masks have been binarized and stored as separate single-class files, then selected only layers that contain relevant organs (spleen, liver, background). Images were resized to dimension of (256, 256, number of slices), all images are labeled with (spleen-background; liver-background). For image normalization we use a Windowing - a known contrast stretching process in which CT image grayscale components of an image are modified to highlight the key anatomy. We scaled the image pixel to [W:180, L:30] which matches liver and abdominal soft tissues.

TABLE I

DATA SETS FOR SPLEEN AND LIVER EXPERIMENTS. EXPLAINING THE DIVISION INTO TRAIN VALIDATION AND TEST (HELD OUT) DATASET. THE SPLIT WAS DONE BY A 60%-15%-25% RULE.

Organ	Partition	# Series	# Images	% Series
Spleen	Training	44	5834	62%
	Validation	10	1144	14%
	Test	17	1875	24%
Liver	Training	78	47658	57%
	Validation	25	11992	18%
	Test	33	12448	24%

4.2. Implementation Details

Case selection was done based on the heuristic acquisition function results (defined in section 4.3.2). In each iteration we ran the model and calculated the entropy and margin on all available training images that we haven't trained on, scores were sorted based on metrics results 10 images were added to the training set of the next iteration, out of the top 20% that showed the highest heuristic acquisition function score.

4.3. Parameters Optimization

In this section we will overview the process of finding the best set of values for the parameters of the framework in order to achieve the best performance in segmentation tasks. The parameters that was set are pool size (subsection 5.3.1), margin distance (subsection 5.3.2) and randomization in image selection (subsection 5.3.3)

4.3.1 Pool size selection

There is a trade-off between training time and pool size selection, smaller pool size will produce a higher mIoU with smaller tagging budget but with more rounds of annotation and training time. We considered {1, 5, 10, 20} images as pool sizes. Since we wanted to produce a model which matches the performance goals but without overhead in training time, the 10 was selected as the optimal sample size.

4.3.2. Margin distance selection (ϵ)

The uncertainty was calculated as the absolute distance of probability from the threshold and set to be smaller than 5%, we tried to optimize between the risk of misclassification and the efficiency of the selection process and prediction results (see Figure. 7). based on grid search ϵ in the range of (0.01-0.1), the performance of the model evaluated using the validation set

$$\text{margin score} = \sum_{j=0}^{256^2} |(p_j - \text{threshold})| < 0.05$$

Equation 6 : Implementation of the margin score

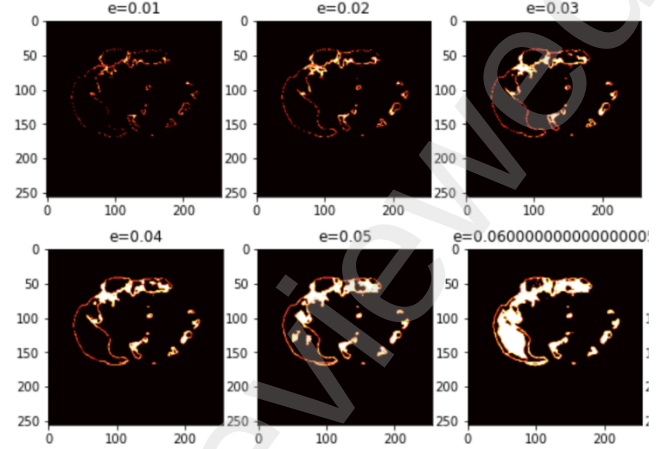


FIGURE 7. Visualization of the ϵ selection process. The figure shows the visualization of offered regions of interest based on the margin score for each image, where the selected ϵ appear above the image, and the series of figures correspond to ϵ in the range of (0.01-0.1) respectively from left to right. From the sequence of images we can see that for that for $\epsilon > 0.05$ we select regions both within the segmented area and in the background, which will lead to high uncertainty in segmentation output, therefore a smaller ϵ was selected in order to restrict roi selection correctly.

4.3.3. Inserting randomization and ranking for image selection

Using AL strategies may suffer from a lack of diversity in case selection, so we decided to reduce it by adding a random factor for the acquisition function. We first rank all the images using the acquisition function, and then sample 10 images uniformly from the top ranked images, denoted by B%. In order to select B we ran the model on different sizes in the range 5%-30% and selected the option that improved the model the most in the next iteration. Since the best results in the experiment were obtained with 20%, thus the B parameter was set accordingly. (Figure 8)

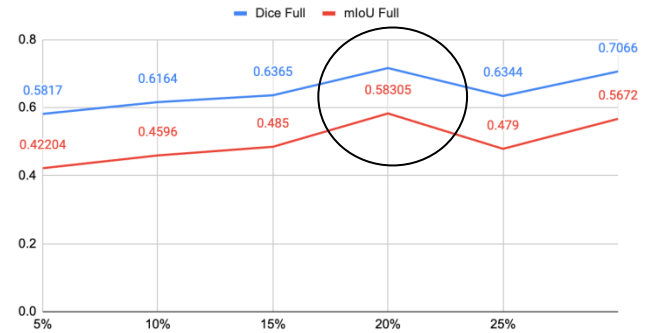


FIGURE 8. Selection process based on the top percentage of images. The percentage parameter denoted B was selected by running the model on different B sizes range (5%-30%) and selecting the option that improved the model the most in the next iteration. The best result was obtained by B= 20% therefore B was set accordingly.

V. RESULTS AND DISCUSSION

Table 2, presents the results obtained on the liver and spleen by the AL process. We compare the two heuristics to the

preliminary baseline and performance goals. The training time and number of images needed for model training (annotation budget) in each phase show the clear advantage of this approach.

TABLE 2:

THE FINAL MODEL RESULTS FROM COMPARISON BY DICE AND mIoU METRICS TRAINING SET SIZE AND TRAINING TIME. BEST RESULTS WERE DISPLAYED FOR SPLEEN WITH CROSS ENTROPY CASE SELECTION AND FOR LIVER WITH MARGIN SCORE. TRAINING TIME AND NUMBER OF IMAGES NEEDED FOR TRAINING (TAGGING BUDGET) WAS REDUCED IN BOTH HEURISTICS.

Model	Training Time (minute)	Number of Tagged images (for Training)	Performance on Masked Held-out set	Performance on Held out set	Tagged Images Reduction (%)
SFull	42	5834	Dice: 0.934 mIoU: 0.875	Dice: 0.907 mIoU: 0.831	-
SInitial	3.3	260	Dice: 0.892 mIoU: 0.805	Dice: 0.611 mIoU: 0.440	4.46
SMargin	36.3	290	Dice: 0.919 mIoU: 0.851	Dice: 0.908 mIoU: 0.833	4.98
SEntropy	34.5	300	Dice: 0.926 mIoU: 0.864	Dice: 0.914 mIoU: 0.842	5.15
LFull	70.3	47658	Dice: 0.838 mIoU: 0.758	Dice: 0.778 mIoU: 0.673	-
LInitial	4	1008	Dice: 0.811 mIoU: 0.718	Dice: 0.621 mIoU: 0.466	2.16
LMargin	62.1	1078	Dice: 0.878 mIoU: 0.826	Dice 0.855 mIoU 0.785	2.27
LEntropy	43.6	1048	Dice: 0.869 mIoU: 0.810	Dice 0.812 mIoU 0.717	2.20

From table 2 we can see that we managed to achieve better results on a full held-out set for both the spleen and liver segmentation tasks and outperform performance goals, which were built based on a training set containing only 2.5% of images available for training (max images needed for training 300 in spleen case and 1078 for liver divided by available for training pool size). For the masked held-out set containing only images with spleen-liver anatomy (removing CT slices containing background only) we managed to outperform only on the liver segmentation task with 0.855 dice score and 0.785 mIoU vs. 0.778 dice score and 0.673 mIoU on full training set. were for the spleen segmentation model on a full training dataset we managed to achieve 0.934 dice score and 0.875 mIoU while our best model score based on the entropy score achieved performance of 0.926 dice score and 0.864 mIoU.

In figure 9 we can see the dice score on each iteration in the AL refinement loop, for cases where the number of iteration is larger than needed (spleen best results were achieved after

4 iteration) we completed with the best result. we can see a drop in performance after 5 iterations in the liver-margin model that can be explained with sampling bias - AL selects the most informative samples, but this can lead to a biased dataset. if the selected sample is not representable of the overall dataset - for example images containing lesions or cancer, the model may overfit to those selected samples and perform poorly on the held-out set.

The purple and brown lines present the average results of the spleen and liver models for each acquisition function and are defined as an aggregative model. we can see that margin performs slightly better than entropy score in the last iteration.

Training time presented as accumulative training time over all iteration, we can see that we arrive faster to better results using Entropy heuristic, but achieve lower results than margin based case selection in the overall segmentation task. Tagging budget we managed to outperform a model with full (both liver and spleen) segmentation with 2% of the training data and 40% of the full dataset (including tags for validation and held-out set). In the liver segmentation task and with 2% of the training set and in the spleen segmentation 5% of the training set images, the gap can be estimated as approximately 700 tagger working hours (for the average tagging time of 40s per image).

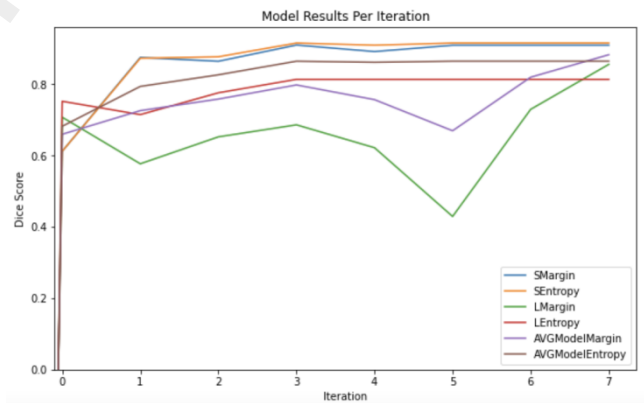


FIGURE 9. Summary of model results in each iteration. Each line represents a different model score represented by anatomy-heuristic for case selection. The purple and brown lines present the average results of the spleen and liver models based on the margin metric.

5.1. Full dataset vs. masked dataset

Both mIoU and Dice were calculated using adjustment by adding smoothing parameters in both the numerator and denominator to avoid cases of division with zero for images that do not contain annotations ($TP = 0$) and successful background predictions ($TP + FP + FN = 0$). This explains the gap when comparing results between full held-out set model results to only masked held-out sets. We can see the

highest gaps appear in models that trained on full training sets for spleen model with gap of 2.6% in dice score and liver model with 6% difference in dice score.

When comparing results on mask vs. full dataset we can see that for liver we outperform in both dataset compared to performance goals defined by the full model.

For spleen, the performance goals on the held-out set was set to 0.908 dice score and 0.833 mIoU after three training iterations with margin score and four iterations with entropy score with 5% of training images. Although we managed to outperform on a full held-out set, we didn't manage to outperform on a masked held-out set which means that the model deals well with FP cases with lower performance on the TP cases. Yet, statistically we can say that models perform the same (the difference $<0.1\%$ of model results is not statistically significant). The reason for not reaching better results with active learning on masked images can be due to (1) the experiments was designed to support the spleen and liver and parameters (e.g. pool size and initial training set size, see explanation in experiment section) are optimized to maximize the overall model performance, (2) since the anatomy of spleen is smaller than liver more images containing spleen should have been selected and in order to deal with data imbalance, similarly for the pool size exposed to model in each iteration should have been smaller than 10. (3) The high performance goal which was set to 0.934 dice and 0.875 for mIoU on the masked held-out set, is difficult to outperform.

5.2. Multi Organ vs. single-organ model

Segmentation solving might suffer from imbalance issues, for example smaller classes in a multi-class segmentation relative to other larger classes and the background (Taghanaki et al. 2019). Since the liver tissue is much larger voxel than spleen tissue we suffer from those issues and our model predicted all voxels as either liver or background results presented in Figure 10.

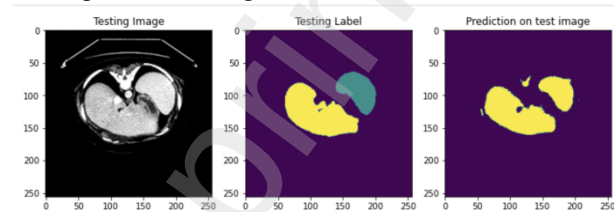


FIGURE 10. Multi class organ segmentation model results. The left image contains the original input CT scan, the center image contains the test label and the rightmost image contains the model prediction. The model was trained on the Synapses dataset containing both liver and spleen annotation. From the right image we can see the model managed to segment both spleen and liver but classified both of them as liver

Also Open source multi organ dataset are hard to find, since we had limitations in tagging budget and most benchmark datasets only provide segmentation masks of a single organ we decided to create a single organ segmentation and to aggregate the result of each pair into a weight-averaged model.

By aggregating the results of the best models, creating a cumulative prediction based on the spleen and liver segmentation maps, using AL with margin heuristic, we manage to achieve a 0.881 avg dice score which are equivalent to the performance goal presented in related researches - 0.88 (Amyar et al, 2020)

Figure 11 presents the results of an aggregated model where segmentation for both spleen and liver appear on the same image scan.

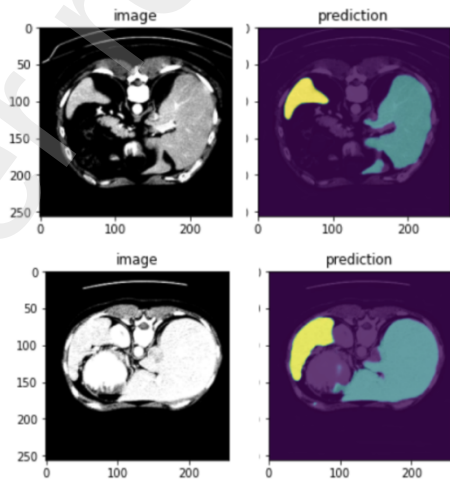


FIGURE 11. Aggregate results to create a dual output with both spleen-liver segmentation. We run inference on all held out sets and then summarize the results. The model selected based on margin score which demonstrated the highest performance results among other heuristics.

5.3. Margin vs. Entropy scores

Comparing the different querying strategies, overall margin score avg result was slightly higher with 0.881 avg dice score and 0.809 mIoU vs. 0.863 avg dice score vs. 0.779 mIoU for entropy score. We can see that major change is for liver segmentation where the gap is 6%. From looking at the training graph (Figure 9) we can see that the entropy curve is smoother with changes while margin score has spikes and major changes throughout the iterations which may indicate the diversity in cases selection. In literature (Settles, 2009) describes the different behavior of those functions, entropy measure does not favor instances where only one of the labels is highly unlikely because the model is fairly certain that it is not the true label. margin on the other hand tends to utilize the information if the model cannot distinguish between the remaining two classes. This can explain the

higher rate of false positives in margin score vs. entropy score.

VI. SUMMARY

In this work, we propose a framework for organ binary segmentation on CT scans, that uses active learning in order to reduce the cost-benefit ratio of the annotation process for DL segmentation models. We suggest an automatic end-to-end framework that effectively training a semantic segmentation model by selecting for labeling images that are highly informative for the task. We focused on implementing the framework for two anatomies spleen and liver and compared two active learning strategies where one strategy is based on the calculation of uncertainty measured by margin score and the other one calculates the entropy score..

Our experiments with multiple datasets and a common 2D Unet architecture demonstrate that our approach achieves comparable performance while reducing the annotation effort, storage capacity and computational efforts.

Furthermore, we demonstrate the robustness of our approach and its adaptability to new organ segmentation tasks.

Future work will include the extension of this approach to 3D segmentation, other anatomies and modalities, and obtaining further generalization of the approach, by expanding it to self supervised segmentation approaches.

Due to the significant role of ground truth annotation in the current ML-DL era, we believe that our research can have a wide impact and contribution on medical imaging and computer vision application in general.

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